

大学化学期末复习配套考点题目 (致新书院)

ZJ (配套课件笔记使用) 专一针对大学化学, 化学原理的朋友也可快速扫描

宗旨是帮大家取得一个理想的分数

期末不涉及期中前内容的考察

Ch 11

Point 1 分子间作用力及各种比较

3. In which of the following molecules is hydrogen bonding likely to be the most significant component of the total intermolecular forces?

- A) CO_2 B) $\text{C}_6\text{H}_{13}\text{NH}_2$ C) CH_4 D) $\text{C}_5\text{H}_{11}\text{OH}$ E) CH_3OH

4. What types of intermolecular forces exist in a mixture of NH_3 and H_2S ?

- A) dispersion forces and hydrogen bonds
B) dispersion forces, dipole-dipole forces, and hydrogen bonds
C) dispersion forces, hydrogen bonds, and ion-dipole forces
D) dispersion forces
E) dispersion forces and dipole-dipole forces

注意氢键对应元素有且只有那三个

5. _____ are particularly polarizable.

- A) Small nonpolar molecules
B) Small polar molecules
C) Large nonpolar molecules
D) Large polar molecules
E) Large molecules, regardless of their polarity

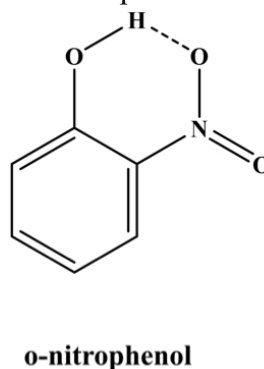
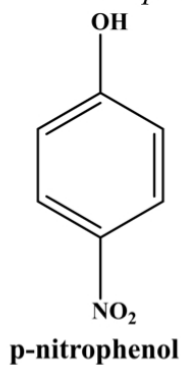
6. Of the following, _____ is the most volatile.

- A) CH_2Br_2 B) CH_4 C) CH_3Br D) CHBr_3 E) CBr_4

7. Which of the following is the correct order of boiling points for BaCl_2 , H_2 , CO , HF and Ne ?

- A) $\text{H}_2 < \text{Ne} < \text{HF} < \text{CO} < \text{BaCl}_2$
B) $\text{H}_2 < \text{Ne} < \text{CO} < \text{HF} < \text{BaCl}_2$
C) $\text{H}_2 < \text{HF} < \text{Ne} < \text{CO} < \text{BaCl}_2$
D) $\text{H}_2 < \text{Ne} < \text{CO} < \text{BaCl}_2 < \text{HF}$
E) $\text{BaCl}_2 < \text{H}_2 < \text{Ne} < \text{CO} < \text{HF}$

18. (5 points) Two isomers of *p*-nitrophenol and *o*-nitrophenol are shown below.

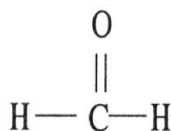


(a) One isomer has a melting point of 44 °C and the other 144 °C. Which isomer has which melting point? (b) Explain why.

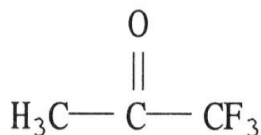
去年有一道小题考了同一个分子式但不同结构式对于分子间作用力大小的影响

5. Which one of the following substances will have hydrogen bonding as one of its intermolecular forces?

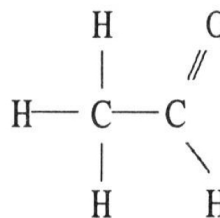
A)



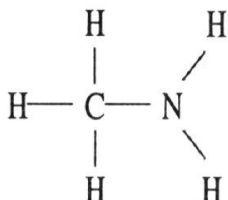
B)



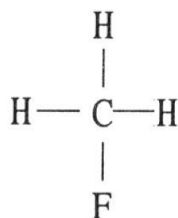
C)



D)



E)



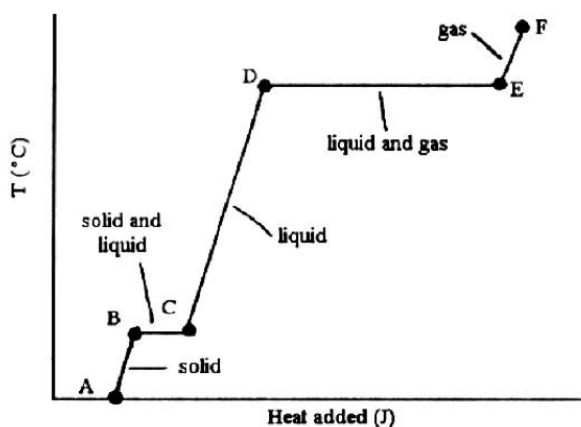
9. Large intermolecular forces in a substance are shown by _____.

- A) low vapor pressure
- B) high boiling point
- C) high heats of fusion and vaporization
- D) high critical temperatures and pressures
- E) all of the above

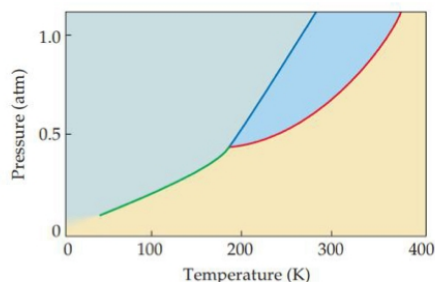


Point 2 相图相关

11. The phase changes $B \rightarrow C$ and $D \rightarrow E$ in the heating curve shown below are not associated with temperature increases because the heat energy is used up to _____.
- A) increase distances between molecules B) break intramolecular bonds
C) rearrange atoms within molecules D) increase the velocity of molecules
E) increase the density of the sample



15. Based on the phase diagram of a hypothetical substance below, what happens to it as it is heated from 50 K to 200 K at 0.25 atm?



- A) It melts at about 125 K.
B) It sublimates at about 125 K.
C) It condenses at about 125 K.
D) It boils at about 125 K.
E) It reaches the triple point at about 200 K.

注意哪个地方是supercritical fluid (去年考过)

Point 3 蒸汽压

13. In general, the vapor pressure of a substance increases as _____ increases.
- A) viscosity B) temperature C) hydrogen bonding
D) molecular weight E) surface tension

12. The vapor pressure of any substance at its normal boiling point is _____.
- A) 1 Pa
B) 1 torr
C) 1 atm
D) equal to atmospheric pressure
E) equal to the vapor pressure of water

注意蒸汽压和气压的区别 (一个是针对溶液, 一个是针对气体)

蒸汽压更像是一种产生气体的能力

Point 4 液晶

14. Molecules with only single bonds do not generally exhibit liquid-crystalline properties because _____.
A) molecules with only single bonds are too big to exhibit liquid-crystalline properties
B) molecules without multiple bonds lack the flexibility necessary for alignment
C) molecules with single bonds are gases
D) molecules without multiple bonds lack the rigidity necessary for alignment
E) molecules without multiple bonds are too small to exhibit liquid-crystalline properties

Ch 13

Point 1 溶剂溶质间的相互作用

2. Indicate the main type of solute-solvent interaction HCl in acetonitrile (CH_3CN).
A) ion-ion forces B) ion-dipole C) London dispersion D) dipole-dipole
E) hydrogen bonding
6. The process of a substance sticking to the surface of another is called _____.
A) absorption
B) diffusion
C) effusion
D) adsorption
E) coagulation
1. Which of the following explanations accounts for the fact that the ion-solvent interaction is greater for Li^+ than for K^+ ?
A) Li^+ is of lower mass than K^+ .
B) The ionization energy of Li is higher than that for K.
C) Li^+ has a smaller ionic radius than K^+ .
D) Li has a lower density than K.
E) Li reacts with water more slowly than K.

Point 2 溶解度（饱和）相关问题

8. Which of the following substances is more likely to dissolve in H_2O ?
A) CCl_4
B) Ar
C) C_6H_6
D) CH_3OH

10. Indicate which statement is false in the formation of a solution.

- A) A solute will dissolve in a solvent if solute-solute interactions are smaller than solute-solvent interactions.
- B) An increase in entropy favors mixing.
- C) In making a solution, the enthalpy of mixing is always a positive number
- D) NaCl dissolves in water but not in benzene because the water-ion interactions are stronger than benzene-ion interactions.

永远是放热

11. The solubility of MnSO_4 monohydrate in water at 20°C is 70.0 g per 100.0 mL of water. A solution at 20°C that is 4.22 M in MnSO_4 monohydrate is best described as a(n) _____ solution. The formula weight of MnSO_4 monohydrate is 168.99 g/mol.

- A) hydrated
- B) solvated
- C) saturated
- D) supersaturated
- E) unsaturated

6. A 5.25% by mass aqueous solution of NaI can be prepared by dissolving _____ .

- A) 5.25 mol NaI in 1.00 L solution
- B) 1.00 g NaI in 5.25 g water
- C) 5.25 g NaI in 94.75 mL water
- D) 5.25 mol NaI in 1.00 kg water
- E) 5.25 g NaI in 100 g water

12. A solution contains 22 ppm of benzene. The density of the solution is 1.00 g/mL. This means that _____.

- A) there are 22 mg of benzene in 1.0 g of this solution
- B) 100 g of the solution contains 22 g of benzene
- C) 1.0 g of the solution contains 22×10^{-6} g of benzene
- D) 1.0 L of the solution contains 22 g of benzene
- E) the solution is 22% by mass of benzene

16. Calculate the molality of a 10.0% (by mass) aqueous solution of hydrochloric acid.

- A) 0.274 m
- B) 2.74 m
- C) 3.04 m
- D) 4.33 m
- E) The density of the solution is needed to solve the problem.

注意到计算过程中各种各样单位的表示

Point 3 溶液的依数性

3. Of the following, a 0.1 M aqueous solution of _____ will have the lowest freezing point.
- A) NaCl
 - B) $\text{Al}(\text{NO}_3)_3$
 - C) K_2CrO_4
 - D) Na_2SO_4
 - E) sucrose

等于在问particles的浓度

7. Seawater contains 3.4 g of salts for every liter of solution. Assuming that the solute consists entirely of NaCl (in fact, over 90% of the salt is indeed NaCl), calculate the osmotic pressure of seawater at 20 °C. ($R = 0.0821 \text{ L} \cdot \text{atm} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$)
- A) 272 atm B) 2.8 atm C) 2.69 atm D) 20.4 atm E) 817 atm

5. The vapor pressure of benzene, C_6H_6 , is 100.0 torr at 26.1 °C. Assuming Raoult's law is obeyed, how many moles of a nonvolatile solute must be added to 100.0 mL of benzene to decrease its vapor pressure by 10.0% at 26.1 °C? The density of benzene is 0.8765 g/cm^3 .
- A) 0.011237 B) 0.11237 C) 0.1248 D) 0.1282 E) 8.765

20. (a) Does a 0.10 m aqueous solution of NaCl have a higher boiling point, a lower boiling point, or the same boiling point as a 0.10 m aqueous solution of $\text{C}_6\text{H}_{12}\text{O}_6$? Explain.
(b) The experimental boiling point of the NaCl solution is lower than that calculated assuming that NaCl is completely dissociated in solution. Why is this the case?

(a) A 0.10 m aqueous solution of NaCl have a **higher** boiling point. Boiling-point elevation is directly related to total moles of dissolved particles. NaCl is a strong electrolyte, The 0.10 m NaCl produces twice as many dissolved particles as 0.10 m $\text{C}_6\text{H}_{12}\text{O}_6$. So its boiling point is higher than that of 0.10 m $\text{C}_6\text{H}_{12}\text{O}_6$.

(b)
In solutions of strong electrolytes like NaCl, **ion pairing** reduces the effective number of particles in solution, decreasing the change in boiling point. The actual boiling point is then lower than the calculated boiling point for a 0.10 m solution.

21. List the following aqueous solutions in order of increasing boiling point: 0.120 m glucose, 0.050 m LiBr, 0.050 m $\text{Zn}(\text{NO}_3)_2$.

该类型为期末必考的简答题排序

Point 4 胶体相关问题

13. Which of the following cannot be a colloid?
- A) an emulsion
 - B) an aerosol
 - C) a homogeneous mixture
 - D) a foam
 - E) All of the above are colloids.

胶体不是溶液

15. An "emulsifying agent" is a compound that helps stabilize a hydrophobic colloid in a hydrophilic solvent (or a hydrophilic colloid in a hydrophobic solvent). Which of the following choices is the best emulsifying agent?
- A) CH_3COOH
 - B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 - C) $\text{CH}_3(\text{CH}_2)_{11}\text{COOH}$
 - D) $\text{CH}_3(\text{CH}_2)_{11}\text{COONa}$

Ch 14

Point 1 反应速率的影响因素和单位

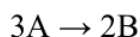
1. A burning splint will burn more vigorously in pure oxygen than in air because _____.

- A) oxygen is a reactant in combustion and concentration of oxygen is higher in pure oxygen than in air
- B) oxygen is a catalyst for combustion
- C) oxygen is a product of combustion
- D) nitrogen is a product of combustion and the system reaches equilibrium at a lower temperature
- E) nitrogen is a reactant in combustion and its low concentration in pure oxygen catalyzes the combustion

2. Of the following, all are valid units for a reaction rate except _____.

- A) mol/L B) M/s C) mol/hr D) g/s E) mol/L-hr

3. Consider the following reaction:



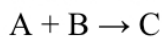
The average rate of appearance of B is given by $\Delta[\text{B}]/\Delta t$. Comparing the rate of appearance of B and the rate of disappearance of A, we get $\Delta[\text{B}]/\Delta t = \text{_____} \times (-\Delta[\text{A}]/\Delta t)$.

- A) -2/3 B) +2/3 C) -3/2 D) +1 E) +3/2

注意正负号

Point 2 反应级数 (以下几乎为必考选择题)

4. The data in the table below were obtained for the reaction:



Experiment Number	[A] (M)	[B] (M)	Initial Rate (M/s)
1	0.451	0.885	1.13
2	0.451	1.77	1.13
3	1.35	0.885	10.17

The rate law for this reaction is rate = _____.

- A) $k[A]^2[B]$ B) $k[A][B]$ C) $k[A]^2$ D) $k[A]^2[B]^2$ E) $k[C]$

5. The elementary reaction $2 \text{NO}_2(g) \rightarrow 2 \text{NO}(g) + \text{O}_2(g)$

is second order in NO_2 and the rate constant at 600 K is $6.77 \times 10^{-1} \text{ M}^{-1} \cdot \text{s}^{-1}$.

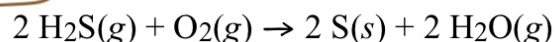
The reaction half-life at this temperature when $[\text{NO}_2]_0 = 0.45 \text{ M}$ is _____ s.

- A) 3.3 B) 1.0 C) 0.98 D) 1.5 E) 0.30

6. The rate constant of a first-order process that has a half-life of 3.50 min is _____ s^{-1}

- A) 0.693 B) 3.30×10^{-2} C) 1.98 D) 0.198 E) 3.30×10^{-3}

3. Which of the following statements is true concerning the reaction given below?



- A) The rate law is $\text{Rate} = k[\text{H}_2\text{S}]^2[\text{O}_2]$.
B) The reaction is second-order in $\text{H}_2\text{S}(g)$ and first-order in $\text{O}_2(g)$.
C) The reaction is first-order in $\text{H}_2\text{S}(g)$ and second-order in $\text{O}_2(g)$.
D) The rate law is $\text{Rate} = k[\text{H}_2\text{S}][\text{O}_2]$.
E) The rate law may be determined only by experiment.

Point 3 rate law factor

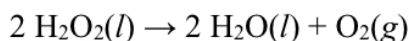
8. The rate of a reaction depends on _____.

- A) collision frequency
B) collision energy
C) collision orientation
D) all of the above
E) none of the above

12. What is the magnitude of k for a first-order reaction at 60.00°C if $E_a = 121 \text{ kJ/mol}$ and rate constant is $1.75 \times 10^{-1} \text{ s}^{-1}$ at 20.00°C ?

- A) 385 B) 136 C) 68 D) 34 E) 17

14. The reaction below is first order in $[\text{H}_2\text{O}_2]$:



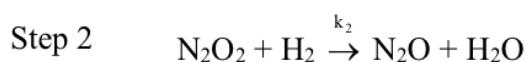
A solution originally at 0.600 M H_2O_2 is found to be 0.075 M after 54 min. The half-life for this reaction is _____ min.

- A) 6.8 B) 18 C) 14 D) 28 E) 54

如果没有那么恰好，常规该怎么做？

Point 4 基元反应及反应机理

11. A possible mechanism for the gas phase reaction of NO and H_2 is as follows:

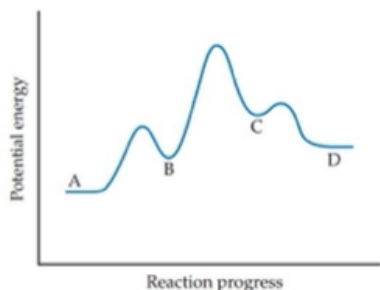


Which of the following statements concerning this mechanism is not directly supported by the information provided?

- A) Step 1 is the rate determining step.
B) N_2O_2 is an intermediate.
C) There is no catalyst in this reaction.
D) The rate law for step 1 is $\text{rate} = k_1[\text{NO}]^2$.
E) All steps are bimolecular reactions.

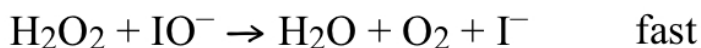
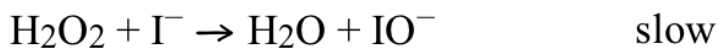
Part B Short questions (Write legibly and show all work for all steps in the problem)

16. (5 points) Consider the following energy profile in the reaction $\text{A} \rightarrow \text{D}$.



(a) How many elementary reactions are in the reaction mechanism? (b) How many intermediates are formed in the reaction $\text{A} \rightarrow \text{D}$? And indicate them. (c) Which step is rate determining? (d) For the overall reaction, is ΔE positive, negative, or zero?

7. Below is a proposed mechanism for the decomposition of H_2O_2 .



Which of the following statements is incorrect?

- A) IO^- is a catalyst.
- B) I^- is a catalyst.
- C) The net reaction is $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$.
- D) The reaction is first-order with respect to $[\text{I}^-]$.
- E) The reaction is first-order with respect to $[\text{H}_2\text{O}_2]$.

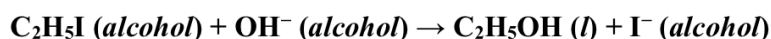
slow/fast的这种如果没有写step1, step2, 默认上面的方程式是step1!!!

18.) Which of the following reactions is not an example of homogeneous catalysis?

- A) $2\text{O}_3(\text{g}) \xrightarrow{\text{Cl} \cdot (\text{g})} 3\text{O}_2(\text{g})$
- B) $\text{S}_2\text{O}_8^{2-}(\text{aq}) + 2\text{I}^-(\text{aq}) \xrightarrow{\text{Fe}^{2+}(\text{aq})} 2\text{SO}_4^{2-}(\text{aq}) + \text{I}_2(\text{aq})$
- C) $2\text{H}_2\text{O}_2(\text{aq}) \xrightarrow{\text{Br}^-(\text{aq})} 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
- D) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \xrightarrow{\text{Fe}(\text{s})} 2\text{NH}_3(\text{g})$
- E) $\text{CH}_3\text{CO}_2\text{CH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \xrightarrow{\text{H}^+(\text{aq})} \text{CH}_3\text{CO}_2\text{H}(\text{aq}) + \text{CH}_3\text{OH}(\text{aq})$

此为催化剂容易忽略的一点

20. (8 points) The reaction between ethyl iodide ($\text{C}_2\text{H}_5\text{I}$) and hydroxide ion in ethanol ($\text{C}_2\text{H}_5\text{OH}$) solution, has an activation energy of 86.8 kJ/mol and a frequency factor (A) of $2.10 \times 10^{11} \text{ M}^{-1}\text{s}^{-1}$.



- (a) Predict the rate constant for the reaction at 40 °C.
- (b) A solution of KOH in ethanol is made up by dissolving 0.343 g KOH in ethanol to form 279.0 mL of solution. Similarly, 1.496 g of $\text{C}_2\text{H}_5\text{I}$ is dissolved in ethanol to form 279.0 mL of solution. Equal volumes of the two solutions are mixed. Assuming the reaction is first order in each reactant, what is the initial rate at 40 °C?
- (c) Assuming the frequency factor and activation energy do not change as a function of temperature, calculate the rate constant for the reaction at 50 °C.

Ch 15

Point 1 平衡常数及相关计算

5.) The equilibrium-constant expression depends on the _____ of the reaction.

- A) stoichiometry and mechanism
- B) mechanism
- C) stoichiometry
- D) the quantities of reactants and products initially present
- E) temperature

14. At 500 K the equilibrium constant for the reaction $2\text{NO}(g) + \text{Cl}_2(g) \rightleftharpoons 2\text{NOCl}(g)$ is $K_p = 52.0$. An equilibrium mixture of the three gases has partial pressures of 0.0950 atm and 0.171 atm for NO and Cl_2 , respectively. What is the partial pressure of NOCl in the mixture?

- A) 0.845 atm B) 2.97×10^{-5} atm C) 0.283 atm D) 8.02×10^{-2} atm E) 5.45×10^{-3} atm

16. Consider the equilibrium that is established in a saturated solution of silver chloride, $\text{Ag}^+(aq) + \text{Cl}^-(aq) \rightleftharpoons \text{AgCl}(s)$. If NaCl solution is added to this solution, what will happen to the concentration of Ag^+ and Cl^- ions in solution?

- A) $[\text{Ag}^+]$ and $[\text{Cl}^-]$ will both increase
 B) $[\text{Ag}^+]$ and $[\text{Cl}^-]$ will both decrease
 C) $[\text{Ag}^+]$ will increase and $[\text{Cl}^-]$ will decrease
 D) $[\text{Ag}^+]$ will decrease and $[\text{Cl}^-]$ will increase
 E) neither $[\text{Ag}^+]$ nor $[\text{Cl}^-]$ will change.

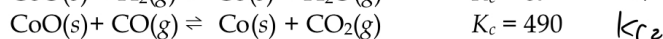
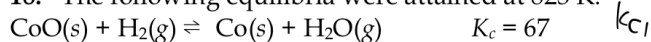
Part B short questions--Write legibly and show all work for all steps in the problem. For mathematical relationships show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (14 points)

17. Consider the following equilibrium, for which $\Delta H < 0$,
 $2\text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g)$.

Of the following changes, predict the equilibrium for the reaction above will shift to right, left or no change.

- (a) $\text{O}_2(g)$ is added to the system.
 (b) the reaction mixture is heated.
 (c) the volume of the reaction vessel is doubled.
 (d) a catalyst is added to the mixture.
 (e) the total pressure of the system is increased by adding a noble gas.
 (f) $\text{SO}_3(g)$ is removed from the system.

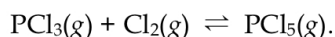
18. The following equilibria were attained at 823 K:



Based on these equilibria, calculate the equilibrium constant for $\text{H}_2(g) + \text{CO}_2(g) \rightleftharpoons \text{CO}(g) + \text{H}_2\text{O}(g)$ at 823 K.

注意 K_p 和 K_c 之间的转换关系，见下面这道题

18. Phosphorus trichloride gas and chlorine gas react to form phosphorus pentachloride gas:



A 7.5-L gas vessel is charged with a mixture of $\text{PCl}_3(g)$ and $\text{Cl}_2(g)$, which is allowed to equilibrate at 450. K. At equilibrium the partial pressures of the three gases are $P[\text{PCl}_3(g)] = 0.124$ atm, $P[\text{Cl}_2(g)] = 0.157$ atm, and $P[\text{PCl}_5(g)] = 1.30$ atm. (a) What is the value of K_p at this temperature? (b) Does the equilibrium favor reactants or products? (c) Calculate K_c for this reaction at 450. K.

3. For the reaction
 $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$
 $K_p = 8.8$ when pressures are measured in atmospheres. Under which of the following conditions will the reaction proceed in the forward direction?
- | $\text{NO}_2(\text{g})$ | $\text{N}_2\text{O}_4(\text{g})$ |
|-------------------------|----------------------------------|
| A) 0.200 atm | 0.352 atm |
| B) 250 mmHg | 400 mmHg |
| C) 0.00255 atm | 0.000134 atm |
| D) 46.5 mmHg | 82.3 mmHg |
| E) 0.138 atm | 0.764 atm |

Point 2 Haber process

2. Which one of the following is true concerning the Haber process?
- A) It is a process used for shifting equilibrium positions to the right for more economical chemical synthesis of a variety of substances.
 - B) It is a process used for the synthesis of ammonia.
 - C) It is another way of stating Le Châtelier's principle.
 - D) It is an industrial synthesis of sodium chloride that was discovered by Karl Haber.
 - E) It is a process for the synthesis of elemental chlorine.

Ch 16

关于后面几章对数有效数字的计算问题:

例如:

$$X = abc \log Y$$

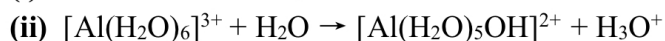
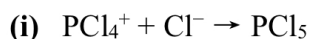
$$Y = abce^X$$

看X的小数点后有几位, 就对应Y有几位有效数字!! (课上详细解释)

Point 1 酸碱定义及共轭酸碱对

1. Which one of the following statements regarding K_w is false?
- A) K_w changes with temperature.
 - B) $\text{p}K_w$ is 14.00 at 25°C .
 - C) The value of K_w is always 1.0×10^{-14} .
 - D) The value of K_w shows that water is a weak acid.
 - E) K_w is known as the ion product of water.
4. Is it possible for a solution to have a negative pH? If so, would that pH signify a basic or acidic solution?
- A) No, the pH range is 1~14.
 - B) No, the definition of pH does not permit it to have a negative value.
 - C) Yes, for any solution with a concentration of base greater than 1 M.
 - D) Yes, for any solution with a concentration of acid greater than 1 M.
 - E) Yes, for any solution with a concentration of acid smaller than 1 M.
9. The dihydrogenphosphate ion, H_2PO_4^- is amphiprotic. In which of the following reactions is/are this ion serving as a Brønsted-Lowry base?
- I. $\text{H}_3\text{O}^+(\text{aq}) + \text{H}_2\text{PO}_4^-(\text{aq}) \rightleftharpoons \text{H}_3\text{PO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 - II. $\text{H}_3\text{O}^+(\text{aq}) + \text{HPO}_4^{2-}(\text{aq}) \rightleftharpoons \text{H}_2\text{PO}_4^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 - III. $\text{H}_3\text{PO}_4(\text{aq}) + \text{HPO}_4^{2-}(\text{aq}) \rightleftharpoons 2 \text{H}_2\text{PO}_4^-(\text{aq})$

11. For each of these reactions, identify the acid among the reactants.



A) Both PCl_4^+ and $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ are Arrhenius acids.

B) Both PCl_4^+ and $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ are Brønsted-Lowry acids.

C) PCl_4^+ is a Arrhenius acid. $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ is a Lewis acid.

D) PCl_4^+ is a Brønsted-Lowry acid. $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ is a Lewis acid.

E) PCl_4^+ is a Lewis acid. $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ is a Brønsted-Lowry acid.

Point 2 酸碱相关计算

5. What is the pOH of an aqueous solution at 25.0 °C that contains $2.5 \times 10^{-4} M$ hydronium ion?

- A) 4.0×10^{-11} B) 10.40 C) 9.40 D) 3.60 E) 7.00

以下两道题非常典型!!!

7. The acid-dissociation constants of sulfurous acid (H_2SO_3) are $K_{a1} = 1.7 \times 10^{-2}$ and $K_{a2} = 6.4 \times 10^{-8}$ at 25.0 °C. Calculate the pH of a 0.163 M aqueous solution of sulfurous acid.

- A) 4.53 B) 1.35 C) 1.86 D) 6.21 E) 1.28

15. What is the pOH of 0.626 M anilinium hydrochloride ($\text{C}_6\text{H}_5\text{NH}_3\text{Cl}$) solution in water at 25 °C, given that K_b for aniline ($\text{C}_6\text{H}_5\text{NH}_2$) is 3.83×10^{-4} ?

- A) 1.811 B) 5.393 C) 8.607 D) 9.393 E) 12.18

Point 3 酸碱性强弱判断

10. In which of the following solutions is the percent ionization of the acid the greatest?

Weak acid	K_a
HBrO	2.5×10^{-9}
$\text{HC}_3\text{H}_5\text{O}_2$	1.3×10^{-5}

A) 0.10 M HBrO(aq)

B) 0.10 M $\text{HC}_3\text{H}_5\text{O}_2$ (aq)

C) 0.025 M HBrO(aq)

D) 0.025 M $\text{HC}_3\text{H}_5\text{O}_2$ (aq)

E) All of the above solutions has the same percent ionization.

13. Of the following substances, the aqueous solutions of _____ is/are basic.

NaHS Cu(NO₃)₂ KHCO₃ NaF

Formula	K_{a1}	K_{a2}
H ₂ S	9.5×10^{-8}	1×10^{-19}
H ₂ CO ₃	4.3×10^{-7}	5.6×10^{-11}

- A) NaHS, Cu(NO₃)₂
 B) KHCO₃, NaHS
 C) NaF only
 D) NaF, KHCO₃
 E) NaHS, KHCO₃ and NaF

8. Which of the following correctly ranks the three monoprotic acids listed in the table below from the weakest to the strongest?

Acid	Concentration	pH
X	0.005 M	2.3
Y	2.0 M	2.8
Z	3.0 M	2.8

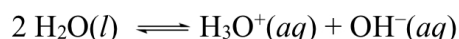
- A) X < Y < Z
 B) X < Z < Y
 C) Y < Z < X
 D) Z < Y < X
 E) Y < X < Z

2. X and Z represent two halogens. A 0.1 M solution of HOX(aq) has a lower pH than a 0.1 M solution of HOZ(aq) has. Which of the following could help explain why?

- A) X has a smaller electronegativity than Z does.
 B) X has a larger electron cloud than Z does.
 C) OX⁻(aq) is less stable than OZ⁻(aq) is.
 D) The O–H bond in HOX(aq) is weaker than that in HOZ(aq).
 E) X has a larger atomic radius than Z does.

B3) Select the stronger base in each of the following pairs: **(a)** BrO⁻ or ClO⁻; **(b)** BrO⁻ or BrO₂⁻; **(c)** HPO₄²⁻ or H₂PO₄⁻; **(d)** F⁻ or Cl⁻.

(b) Based on the change in K_w with temperature, predict whether ΔH is positive, negative, or zero for the autoionization reaction of water and explain:



18. (4 points) The following observations are made about a diprotic acid H₂A:

(a) A 0.10 M solution of H₂A has pH = 3.30.

(b) A 0.10 M solution of the salt NaA is acidic.

Which of the following could be the value of $\text{p}K_{a2}$ for H₂A: (i) 3.22, (ii) 5.30, (iii) 7.47 or (iv) 9.82? Justify your answer.

Ch 17

Point 1 Buffer

2. A buffer contains a weak acid, HA, and its conjugate base. The weak acid has a pK_a of 4.5, and the buffer has a pH of 4.3. Without doing a calculation, state which of these possibilities are correct.

- A) $[HA] = [A^-]$ B) $[HA] > [A^-]$ C) $[HA] < [A^-]$

14. Which one of the following pairs can be mixed together to form a buffer solution?

A) $HONH_2$, $HONH_3Cl$

B) KNO_3 , HNO_3

C) $RbOH$, HBr

D) $NaCl$, HCl

E) H_2SO_4 , K_2SO_4

Point 2 滴定及酸碱指示剂

1. Choose an indicator from the table below for the titration of a 0.100 M solution of a weak acid HA ($K_a=6.2\times 10^{-6}$) with 0.100 M NaOH solution at 298 K.

Indicator name	pH range of color change
Methyl violet	0.0-1.6
thymolphthalein	9.4-10.6
litmus	5.0-8.0
Bromothymol blue	6.0-7.6
phenolphthalein	8.2-10.0

A) Methyl violet

B) thymolphthalein

C) litmus

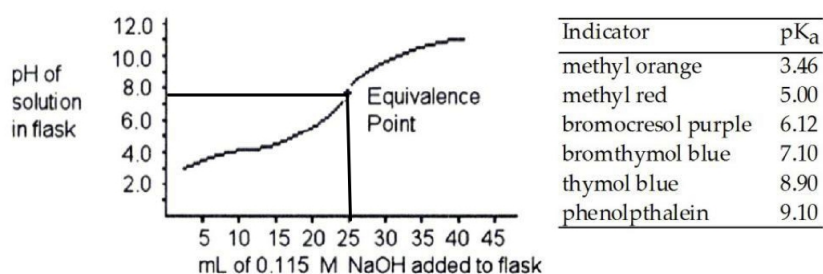
D) Bromothymol blue

E) phenolphthalein

6. A 25.0 mL sample of a solution of a monoprotic acid is titrated with a 0.115 M NaOH solution. The titration curve below was obtained. Which of the following indicators would be best for this titration?

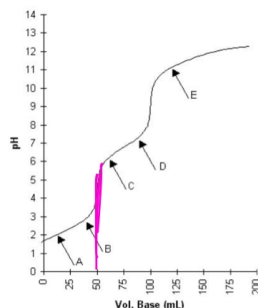
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Chapter 17 Homework



- A) methyl red
- B) bromocresol purple
- C) thymol blue
- D) bromthymol blue
- E) phenolphthalein

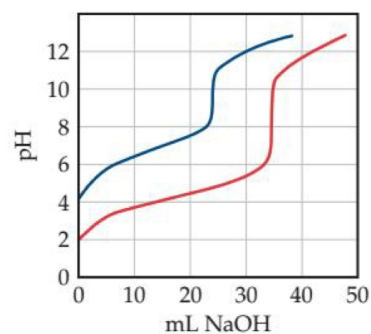
9. At what point in this titration of arsenic acid (H_3AsO_4) with base would one expect to find a mixture of primarily H_2AsO_4^- and a lesser amount of HAsO_4^{2-} ?



- A) Point A on the curve
- B) Point E on the curve
- C) Point B on the curve
- D) Point C on the curve
- E) Point D on the curve

4. Equal volumes of two acids are titrated with 0.10 M NaOH resulting in the two titration curves shown in the following figure. Which curve corresponds to the more concentrated acid solution? Which corresponds to the acid with the larger K_a ?

- A) red, red
- B) blue, red
- C) blue, blue
- D) red, blue



14. Calculate the concentration of the lactate ion in a solution that is 0.080 M in HCl and 0.100 M in lactic acid ($\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, $\text{pK}_a = 3.86$).

- A) 4.83 M B) 0.0800 M C) 7.3×10^{-3} M
D) 3.65×10^{-3} M E) 1.73×10^{-4} M

Point 3 Ksp

Mass solubility / Molar solubility的区别

15. Calculate the solubility in mol/L of $\text{Cu}(\text{OH})_2$ at $\text{pH} = 7.0$. $K_{sp} = 2.2 \times 10^{-20}$.

- A) 5.5×10^{-7} B) 1.77×10^{-7} C) 4.4×10^{-6} D) 2.2×10^{-6} E) 2.1×10^{-4}

18. Use values of $K_{sp}=8.3 \times 10^{-17}$ for AgI and $K_f=1.0 \times 10^{21}$ for $\text{Ag}(\text{CN})_2^-$ to (a) calculate the molar solubility of AgI in pure water, (b) calculate the equilibrium constant for the reaction $\text{AgI}(s) + 2\text{CN}^-(aq) \rightleftharpoons \text{Ag}(\text{CN})_2^-(aq) + \text{I}^-(aq)$, (c) determine the molar solubility of AgI in a 0.100 M NaCN solution.

15. Which compound listed below has the smallest molar solubility in water?

- A) BaCO_3 $K_{sp}=5.0 \times 10^{-9}$
B) $\text{Ca}_3(\text{PO}_4)_2$ $K_{sp}=2.0 \times 10^{-29}$
C) $\text{Cr}(\text{OH})_3$ $K_{sp}=6.7 \times 10^{-31}$
D) AgI $K_{sp}=8.3 \times 10^{-17}$
E) PbF_2 $K_{sp}=3.6 \times 10^{-8}$

21. A sample of 1.25 L of HCl gas at 25 °C and 0.950 atm is bubbled through 0.500 L of 0.150 M NH_3 solution. Calculate the pH of the resulting solution assuming that all the HCl dissolves and that the volume of the solution remains 0.500 L. ($pV=nRT$, $K_b(\text{NH}_3)=1.8 \times 10^{-5}$)

Ch 19

Point 1 Molecular Scale

1. What is the entropy of a system that has only a single microstate?

- A) $S = 0$ B) $S = 1$ C) $S > 1$ D) $S < 1$ E) $S < 0$

12. With thermodynamics, one cannot determine _____.

- A) the speed of a reaction
B) the direction of a spontaneous reaction
C) the extent of a reaction
D) the value of the equilibrium constant
E) the temperature at which a reaction will be spontaneous

Point 2 热力学第二定律

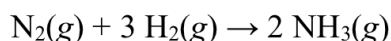
3. A reversible process is one that _____.
A) can be reversed with no net change in either system or surroundings
B) happens spontaneously
C) is spontaneous in both directions
D) must be carried out at low temperature
E) must be carried out at high temperature

定义

4. Which one of the following is always positive when a spontaneous process occurs?
A) $\Delta S_{\text{surroundings}}$ B) ΔS_{system} C) $\Delta S_{\text{universe}}$ D) ΔH_{system} E) $\Delta H_{\text{surroundings}}$
8. The normal boiling point of water is 100.0 °C and its molar enthalpy of vaporization is 40.67 kJ/mol. What is the change in entropy in the system in J/K when 39.3 grams of steam at 1 atm condenses to a liquid at the normal boiling point?
A) -40.7 B) 88.8 C) 373 D) 238 E) -238

Point 3 标准熵和吉普斯自由能

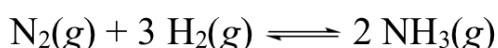
6. In the Haber process, ammonia is synthesized from nitrogen and hydrogen:



ΔG° at 298 K for this reaction is -33.3 kJ/mol. The value of ΔG at 298 K for a reaction mixture that consists of 1.3 atm N_2 , 2.3 atm H_2 , and 0.85 atm NH_3 is _____ kJ/mol.

- A) -3.15×10^3 B) -7.68×10^3 C) -1.5 D) -40.9 E) -108.8

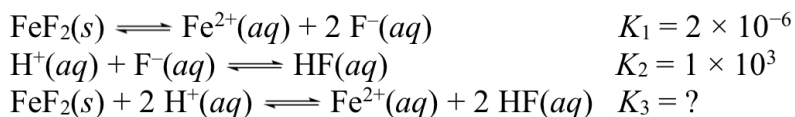
9. The equilibrium constant for the following reaction is 7×10^5 at 25 °C.



The value of ΔG° for this reaction (at 298K) is _____ kJ/mol.

- A) 48.0 B) -48.0 C) 33.3 D) -33.3 E) -22.0

12. On the basis of the information below, the dissolution of $\text{FeF}_2(s)$ in acidic solution is _____.



- A) thermodynamically favorable, because $K_2 > 1$
B) thermodynamically favorable, because $K_3 > 1$
C) not thermodynamically favorable, because $K_1 < 1$
D) not thermodynamically favorable, because $K_3 < 1$
E) Insufficient data are given.

14. The standard Gibbs free energy of formation of _____ is zero.

- (i) C(*graphite*) (ii) Na(s) (iii) Br₂(l)

- A) (i) only
B) (ii) only
C) (iii) only
D) (ii) and (iii)
E) (i), (ii), and (iii)

6. The standard Gibbs free energy of formation of _____ is zero.

- (a) H₂O(l) (b) O(g) (c) Cl₂(g)

- A) (a) only
B) (b) only
C) (c) only
D) (b) and (c)
E) (a), (b), and (c)

期末专块

去年记得的大题相关回顾：

1. 有关于该式子应用的题目：

$$\Delta G = \Delta H - T \Delta S$$

给出一系列属于某个类别的物质，以及他们液态和气态的H，S，分别计算它们的沸点

并判断它们沸点是否处于一个温度区间内（98-105）？

其中不是的让你解释出理由，比如为什么要低或者高一些

去年某个物质的解释是因为存在氢键。

这样可以结合到分子间作用力和熵以及相变相关的综合考察，不难但很繁琐。

2. 有化学反应速率的题目，给出了化学反应式子，但是问环境酸碱度PH（H离子）变化对化学反应平衡的影响？

即需要分析H离子和化学反应式中某个物质（反应物或者生成物）相关作用

该大题其他小问即正常的平衡常数，rate law等的计算

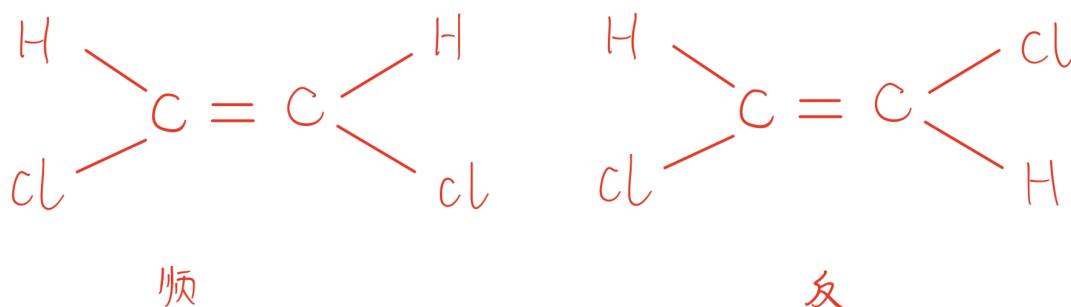
3. 最后一道题就是纯粹的酸碱度加平衡加吉布斯自由能的计算题，相对较为简单

选择题相关回忆 (意外版)

1. 相图中超过了critical point后那个状态名称----Supercritical fluid

2. 某一个物质顺式和反式结构对分子间作用力的影响

记得是这个物质:



简答题:

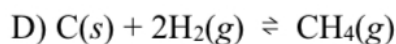
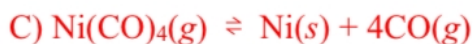
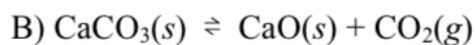
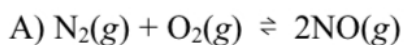
1. 后续题目有相应的

2. 如果给出很复杂的一个物质 (例如画出结构式) 要求能判断它的共轭酸和共轭碱 (画图做答)

3. buffer (包括物质选择和公式计算)

最后的题目:

11. For which of the following reactions is the ratio K_p/K_c largest at 300 K?



14. (7 pts) As the intermolecular attractive forces between molecules increase in magnitude, do you expect each of the following to increase or decrease in magnitude? (a) vapor pressure, (b) heat of vaporization, (c) boiling point, (d) freezing point, (e) viscosity, (f) surface tension, (g) critical temperature.

A3) The first law of thermodynamics can be given as _____.

A) $\Delta E = q + w$

B) $\Delta H_{\text{rxn}}^\circ = \sum n \Delta H_f^\circ (\text{products}) - \sum m \Delta H_f^\circ (\text{reactants})$

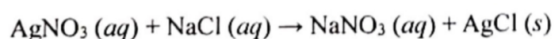
C) for any spontaneous process, the entropy of the universe increases

D) the entropy of a pure crystalline substance at absolute zero is zero

E) $\Delta S = q_{\text{rev}}/T$ at constant temperature

C2) The value of K_a for nitrous acid (HNO_2) at 25 °C is 4.5×10^{-4} . (a) Write the chemical equation for the equilibrium that corresponds to K_a . (b) By using the value of K_a , calculate ΔG° for the dissociation of nitrous acid in aqueous solution. (c) What is the value of ΔG at equilibrium? (d) What is the value of ΔG when $[\text{H}^+] = 5.0 \times 10^{-2} \text{ M}$, $[\text{NO}_2^-] = 6.0 \times 10^{-4} \text{ M}$, and $[\text{HNO}_2] = 0.20 \text{ M}$? Indicate if the forward reaction is spontaneous at 298 K.

C3) The precipitation reaction between $\text{AgNO}_3 (aq)$ and $\text{NaCl} (aq)$ proceeds as follows at 25°C :



(a) Write the net ionic equation of this reaction. (b) Given the data in the table below, calculate ΔH° for the net ionic equation of this reaction. (c) What would you expect for the value of ΔH° of the overall molecular equation compared to that for the net ionic equation? Explain. (d) Use the results from (b) and (c) along with data the table below to determine the value of ΔH_f° for $\text{AgNO}_3 (aq)$.

Compound	$\text{AgNO}_3 (s)$	$\text{Ag}^+ (aq)$	$\text{AgCl} (s)$	$\text{Cl}^- (aq)$	$\text{NaCl} (aq)$	$\text{NaNO}_3 (aq)$
$\Delta H_f^\circ (\text{kJ/mol})$	-124.4	105.90	-127.0	-167.2	-407.1	-446.2

(a) Write the chemical equation that defines the normal boiling point of liquid carbon tetrachloride, $\text{CCl}_4(l)$.

(b) What is the value of ΔG° for the equilibrium in part (a)?

(c) Use data from Appendix C to estimate the normal boiling point of CCl_4 .

Substance	$\Delta H_f^\circ (\text{kJ/mol})$	$\Delta G_f^\circ (\text{kJ/mol})$	$S^\circ (\text{J/mol-K})$
$\text{CCl}_4(g)$	-106.7	-64.0	309.4
$\text{CCl}_4(l)$	-139.3	-68.6	214.4

有效数字举例子:

$$\begin{aligned}\Delta G^\circ &= -9.11 \text{ kJ/mol} \\ \Delta G^\circ &= -RT \ln K \\ \ln K &= \frac{\Delta G^\circ}{-RT} = \frac{-9.1 \times 10^3}{-8.314 \times 298} = 3.672950776 \\ &= 3.7 \\ K &= e^{3.7} = 39.3679005 \\ &= 4 \times 10^1\end{aligned}$$